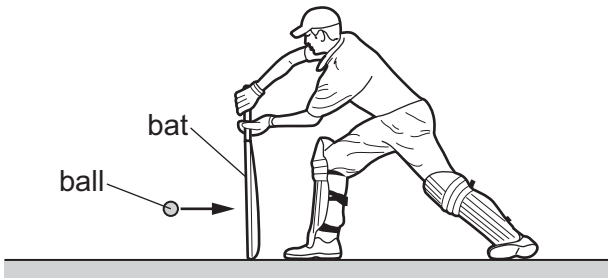
7 A cricket ball of mass 0.15 kg is moving at a speed of 30 m/s.

The diagram shows a batsman holding a bat stationary as the ball hits the bat and bounces back at the same speed in the opposite direction.

The ball is in contact with the bat for 1.0×10^{-3} s.



The table shows the change in momentum of the ball and the magnitude of the force applied to the ball.

Which row is correct?

	$\frac{\text{change of momentum}}{\text{kg m/s}}$	force applied to ball / N
A	4.5	4500
B	4.5	9000
C	9.0	4500
D	9.0	9000